Advanced · AI system architecture · Highlights · IBM Technology

Generative vs Agentic AI: One Call versus a Loop

Understand what actually changes when a model stops answering and starts acting.

For engineers deciding whether a feature needs a model call or an agent · 26 July 2026

The one- or two-page version: what matters, the topics it covers, and every term defined. Read the full lesson for the explanations and worked examples.

Watch the source video · Read the full lesson · Read the condensed version

What matters

- ✓ Generative and agentic AI differ in control flow, not in the model — one is a call, the other is a loop containing that call.
- ✓ A generative system is reactive: it waits for a prompt, produces content, and stops.
- ✓ An agentic system is proactive: given a goal, it perceives, decides, acts, and learns from the result, repeating with minimal human input.
- ✓ The same LLM serves as the reasoning engine in both; in an agent it decides the next action and interprets the last observation.
- ✓ Chain-of-thought decomposition is how an agent turns a vague goal into an ordered sequence of executable steps.
- ✓ Generative workflows keep a human curating every output; agentic workflows spend human attention on the goal and the guard rails instead.
- ✓ A loop has no natural stopping point, so termination conditions, step budgets, and failure handling are your responsibility, not the model's.
- ✓ Cost and latency become unbounded and data-dependent the moment you move from one call to a loop.

What it covers

01 **Reactive generation versus proactive goal pursuit** `0:00`

One produces content and stops. The other pursues an objective across many actions. The gap between them is a while loop.

02 **What a generative model is actually doing** `0:40`

Sophisticated pattern completion: statistical relationships learned from a large corpus, used to predict what plausibly comes next.

03 **The agent loop: perceive, decide, act, learn** `1:22`

Four stages, repeated. Everything that makes an agent an agent lives in this cycle rather than in the model.

04 **The shared substrate: an LLM as reasoning engine** `2:03`

Both approaches usually rest on the same model. In an agent, its job is narrower than people assume: choose the next action, interpret the last result.

05 **Generative in practice: the human curates** `3:28`

Generative workflows produce possibilities. A person selects, corrects and directs — and that review step is load-bearing.

06 **Agentic in practice: multi-step work under a single goal** `4:10`

Agents earn their complexity on tasks that need ongoing management across many steps, where waiting for a human between each one defeats the purpose.

07 **Chain-of-thought: the internal dialogue before the action** `5:38`

The agent breaks a complex goal into smaller logical steps by generating its own reasoning, then acts on the result.

08 **Systems that know when to generate and when to act** `6:20`

The strongest systems are neither purely generative nor purely agentic — they explore options through generation and commit to them through action.

Glossary

Generative AI — A system that produces content in response to a prompt and then stops. Reactive by construction: one request, one result.

Agentic AI — A system that pursues a goal across multiple self-directed actions, observing results and adjusting, with minimal human intervention.

Agent loop — The repeating cycle of perceive, decide, act and learn that defines an agent. The model supplies the decide step; the rest is ordinary software.

Tool (function calling) — A function exposed to the model with a name, description and parameter schema. Tools are the only way an agent can observe or change anything outside its context.

Chain-of-thought reasoning — Generating intermediate reasoning steps before answering or acting, so a complex goal is broken into smaller logical pieces.

Next-token prediction — The mechanism behind text generation: produce a probability distribution over the vocabulary and sample the next token, repeatedly.

Diffusion model — The architecture behind most image and audio generation, which produces output by iteratively denoising random noise.

Step budget — A hard cap on how many iterations an agent may run. Necessary because nothing in the model guarantees the loop terminates.

Message history — The accumulated record of instructions, model replies and tool results. Since the model is stateless, this list is the agent's entire memory.

Human in the loop — A required approval point before a consequential action proceeds. In generative workflows it is continuous; in agentic ones it is placed at specific gates.

Reason–act trace — An execution log alternating stated reasoning with the action taken, used to make an agent's behaviour inspectable after the fact.

Temperature — A sampling parameter controlling how much probability mass outside the top candidate is used. Higher values give more varied output.